2.10.52

EE25BTECH11023 - Venkata Sai

Question:
Let $\mathbf{a} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{c} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$. A vector in the plane of $\mathbf{a}$ and $\mathbf{b}$ whose projection on $\mathbf{c}$ is $\frac{1}{\sqrt{3}}$, is
Solution:
Given

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}. \tag{1}$$

Let $\mathbf{v}$ be the vector in the plane of $\mathbf{a}$ and $\mathbf{b}$

$$[\mathbf{v} \quad \mathbf{a} \quad \mathbf{b}] = 0 \implies \mathbf{v} = \alpha\mathbf{a} + \beta\mathbf{b} \tag{2}$$

$$\mathbf{v} = \alpha \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha + \beta \\ 2\alpha - \beta \\ \alpha + \beta \end{pmatrix}. \tag{3}$$

Mathematically, the projection of $\mathbf{v}$ on $\mathbf{c}$ is defined as

$$\mathbf{d} = k\mathbf{c}, \text{such that} \, (\mathbf{v} - \mathbf{d})^\top \mathbf{d} = 0 \tag{4}$$

yielding

$$(\mathbf{v} - k\mathbf{c})^\top \mathbf{d} = 0 \tag{5}$$

$$\text{or}, k = \frac{\mathbf{v}^\top \mathbf{c}}{||\mathbf{c}||^2}, \implies \mathbf{d} = \frac{\mathbf{v}^\top \mathbf{c}}{||\mathbf{c}||^2} \mathbf{c}. \tag{6}$$

$$||\mathbf{c}||^2 = \mathbf{c}^\top \mathbf{c} = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \tag{7}$$

$$= 1 + 1 + 1 = 3 \implies ||\mathbf{c}|| = \sqrt{3} \tag{8}$$

$$\mathbf{v}^\top \mathbf{c} = \begin{pmatrix} \alpha + \beta & 2\alpha - \beta & \alpha + \beta \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \tag{9}$$

$$= \alpha + \beta + 2\alpha - \beta - \alpha - \beta = 2\alpha - \beta \tag{10}$$

$$\mathbf{d} = \frac{2\alpha - \beta}{3} \mathbf{c} \tag{11}$$

Given $||d|| = \frac{1}{\sqrt{3}}$, Then

$$\frac{|2\alpha - \beta|}{3}||\mathbf{c}|| = \frac{|2\alpha - \beta|}{3}\sqrt{3} = \frac{1}{\sqrt{3}} \implies |2\alpha - \beta| = 1 \tag{12}$$

From equation(3) we know that that x and z coordinates in our required vector must be equal.
If $2\alpha - \beta$ = -1,from the most reliable option i.e option(1) which has the same x and z coordinates, we get

$$\alpha + \beta = 4 \text{ and } 2\alpha - \beta = -1 \tag{13}$$

On adding gives

$$3\alpha = 3 \implies \alpha = 1 \text{ and } \beta = 3 \tag{14}$$

$$\mathbf{v} = \begin{pmatrix} 4 \\ -1 \\ 4 \end{pmatrix} \tag{15}$$

If $2\alpha - \beta$ =1,from the options x and z coordinates are not equal,No option satisfy.

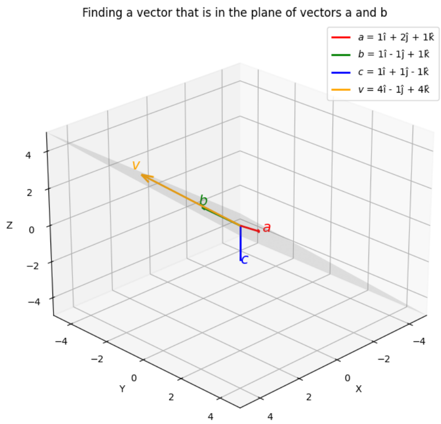

Fig. 0.1